

An Introduction to Reinforcement Learning

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Abstract

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Keywords: Q-learning, Reinforcement learning, State, Agent, Action, TD learning

1. Introduction

1.1. History of Reinforcement learning

Reinforcement learning is a framework, built for understanding how automated agents optimize decision processing actions through interaction of their environment. The development of this process reflects decades of converging ideas that stem from early behavioural psychology research, computational theory, computational neuroscience and machine learning, cultivated to reflect automated learning by reward and predications. Over the years, reinforcement learning has evolved from the basis of theoretical psychology principles into a mathematically driven and biologically grounded paradigm. Today, becoming a widely renown algorithmic method used in central field to study brain function and training artificial intelligence (Sutton & Barto, 2020).

1.1.1. Early Foundations in Psychology

The foundation of reinforcement learning in machine learning can be traced back to the early work of behavioural psychology. Behaviourist Edward Thorndike (1911) is often accredited for his work with animal learning and

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research on learning through trial-and-error learning; where he observed how cats, placed in puzzle boxes show improved escaping performance each time they were repeated placed back into the box. Thorndike proposed the Law of Effect, which suggests that behaviours followed by feelings of satisfaction were more likely to be repeated (Thorndike, 1898; Sutton & Barto, 2020). This principle was further elaborated upon by behaviorist B.F. Skinner, whose theory of operant conditioning emphasized a systematic role of reinforcement and punishment conditions in shaping behaviour (Skinner, 1938). At this time, research was categorically more descriptive and lacked specificity to how they processed occurred. However, their research had introduced the overall premise that learning is driven by feedback contingencies.

1.1.2. Early Computational and Theoretical Developments

By the mid-twentieth century, reinforcement learning became more formally defined with the introduction of computational models of decision making. Richard Bellman and fellow colleagues’ work on dynamic programming provided various ways to describe how optimal decision can be made by considering future outcomes (predictions) (Bellman, 1957; Sutton & Barto, 2020). His formulation of the Bellman equation offered a way to evaluate actions based on long-term consequences.

With the development of the Bellman equation, Bellman and Ronald Howard found that the equation and dynamic programming needed to be utilized under a singular method, introducing the Markov Decision Process (MDP). This mathematical process offered a formal structure to describe situations where decisions are made step by step over time, while also allowing the application of dynamic programming such as policy iteration and value iteration (Sutton & Barto, 2020; Suf, 2025). These early formulations and elements became underlying functions to what would later become reinforcement learning.

Around this time, early models of machine learning were developing in neuroscience and psychology. The work of Arthur Samuel, often considered one of the first examples of machine learning, created a checkers program that improved with experiences (Samual, 1959). Furthermore, shortly after the work of Donald Hebb in 1949, when he proposed that connections between neurons strengthens when they are activated together (Hebb, 1949), Bruse Farley and Wesley Clark developed one of the earliest computational models of learning using artificial neural networks (Farley & Clark, 1954) however, their shift away from trail-and-error learning created confusion on the distinction between what would later be reinforcement learning and supervised learning . As such in the 1960s and 1970s, research around trial-and-error learning became limited (Sutton & Barto, 2020). With some notable exceptions such as John Andreae (1963) STeLLA system and coined the term “learning from a teacher,” and Donald Michie (1961, 1963) use of trial-and-error system to learning tic-tac-toe (Sutton & Barto, 2020).

1.1.3. Temporal Difference and Dopamine

Prediction errors, defined as the difference between expected and unexpected outcomes, involved with the learning process became another major influence towards reinforcement learning. Richard Sutton developed of temporal difference (TD) learning, where predictions are updated gradually over time and allowed continuous learning (Sutton, 1988). In TD learning, predictions are updated step by step using small error that occur each moment. This allows an agent to adjust its expectations and subsequently its actions, making learning more flexible. This became important when neuroscience research observed similar behaviours in the brain to how the prediction error was used in TD learning.

Neuroscience studies showed that dopamine neurons respond in ways that closely match prediction errors (Schultz et al., 1997). For example, some studies showed increase activity of dopamine neurons for unexpected rewards, decrease activity when expected rewards were omitted (getting an undesired outcome) and a shift in activity as the cues for the reward becomes more predictable (Schultz et al., 1997; Mirenowicz & Schultz, 1994; Bayer & Glimcher, 2005). Further evidence in humans showed that the dopamine-related signals help updated the value of actions and guide decision-making (Pessiglione et al., 2006). Together these findings suggested that the in the brain may use a similar process to TD learning to adjust behaviour.

Closely related to this was the development of actor-critic models. A paper from Widrow, Gupta, and Maita (1973) would publish a modified algorithm of Widrow and Hoff's Least-Mean-Square (LMS) to produce a new form of reinforcement learning principle, "selective bootstrap adaptation," that introduced "learning from a critic" (Sutton & Barto, 2015). In this framework, one part of the system selects actions, while another evaluates said actions by value estimate. The critic generates a prediction error signal, which is used to update both components over time. What makes this unique to the previous "learning from a teacher," is that the critic does not need know the outcome to make these predictions, providing a similar pattern of simulation to dopamine neurons (Schultz et al., 1997; Bayer & Glimcher, 2005; Sutton & Barto, 2020).

1.1.4. Reinforcement Learning in Machine Learning

As reinforcement learning became more widely used in computer science, many algorithms developed that could learn directly from experience. One of the most significant algorithms developed was Christopher Watkins' Q-learning, later formalized with Peter Dayan (Watkins & Dayan, 1992). Q-learning built on earlier work in dynamic programming, temporal difference learning, and the MDP framework, as before then, dynamic programming was used to find the optimal solution in MDP. However, the issue with this was that dynamic programming struggled with large pre-set search spaces to find the optimal solution. Q-learning removed this limitation because rather than requiring knowledge about how the environment works, it allows an agent to learn the value of actions through trial-and-error. In Q-learning, the agent learns Q-values, which represent how well a particular action is in a given situation. These values are

than updated over time using prediction errors (Sutton, 1997; Sutton & Barto, 2020; Suf, 2025).

Historically, Q-learning marked a shift toward more flexible and practical reinforcement learning method. It became widely used in the 1990s and early 2000s in fields such as robotics and game play development. Robert Sutton and Andrew Barto also helped bring many of these ideas into a general framework for reinforcement learning (Sutton & Barto, 2020; Suf, 2025).

1.1.5. Modern expansion of Reinforcement Learning Algorithms

Current Reinforcement learning research has expanded past traditional environments into more complex settings. One major direction involves the collaboration of reinforcement learning with large language models through reinforcement learning from human feedback (RLHF), where systems learn from human preferences rather than fixed rewards (Ouyang et al., 2022). Related work in preference-based reinforcement learning also allows agents to learn from comparisons and subjective judgements instead of explicit numerical outcomes, making learning more flexible (Christiano et al., 2017). Reinforcement learning is also being applied in healthcare, where its used to optimize treatment decisions over time by adapting to patient responses (Gottesman et al., 2019). In robotics and autonomous systems, reinforcement learning continues to be used as a tool to improve real-work decision-making and motor control, with a focus on making learning more efficient and adaptable (Kober et al., 2013).

Furthermore, reinforcement learning has advanced quickly with the development of deep learning. Researcher at DeepMind had shown that reinforcement system could learn directly from complex inputs such as images (Mnih et al., 2015). One well-known example is AlphaGo and its successor AlphaGo Zero, which was able to independently learn the game Go and defeat the world champion Lee Sedol in 2016 (Silver et al., 2016). These developments built directly on earlier method of Q-learning and actor-critic models.

At the same time, reinforcement learning continues to be studied in psychology and neuroscience, Researchers use reinforcement models to study decision-making, habits, and disorders such as addiction (Glimcher, 2011). With differences between types of learning, such as model-free and model-based learning, have also been linked to different brain systems (Daw et al., 2005).

1.2. Importance and Applications

- Arshia + some Catherine applications stuff

1.2.1. Applications

Reinforcement learning has been successfully applied to a wide range of domains. Sutton and Barto (2020) explain that an agent learns to interact with an environment through trial and error, in order to achieve long-term goals under uncertainty. The agent's actions influence both immediate outcomes and future states, requiring decisions that account for delayed consequences and involve

planning. At the same time, because outcomes are uncertain, the agent must continuously monitor the environment and adjust its behaviour. Over time, the agent improves its performance through experience, updating its decision-making strategies (Sutton & Barto, 2020).

Robotics is one of the most widely studied domains of reinforcement learning, with applications in manufacturing, logistics, healthcare, and space (Li, 2019). Reinforcement learning enables robots to learn complex sequential behaviors through interaction with their environment, making it well-suited for tasks that involve uncertainty and dynamic conditions (Li, 2019). According to Kober et al. (2013), representations, prior knowledge, and approximate models are fundamental components in robotics. Representations refer to state-action discretization, function approximation, and pre-structured policies, and they aim to make complex reinforcement learning problems tractable. Prior knowledge is crucial in guiding the learning process and improving efficiency, including initial policies, demonstration, task structuring, and constraints on actions or policies. Approximate models also improve learning efficiency by enabling simulation and prediction of environment dynamics (Kober et al., 2013). However, Li (2019) states that discrepancies between simulation and the real world, referred to as the sim-to-real gap, might be a challenge. It could be addressed by improving the simulator or designing policies such as randomization in dynamics, observations, and environmental conditions (Li, 2019). In addition, Akalin and Loutfi (2021) explain that the development of social robots has introduced new forms of reward mechanisms. In interactive reinforcement learning, robots learn from human feedback, which may be explicit (e.g., ratings or labels) or implicit (e.g., gestures, speech, or emotional cues).

Reinforcement learning enables self-driving cars to learn optimal driving policies by rewarding desired driving behaviors and penalizing errors, allowing agents to learn through trial and error and ultimately reach the destination (Sivamayil et al., 2023). According to Badue et al. (2021), the self-driving system is typically divided into the perception system and the decision-making system. The perception system is responsible for interpreting data from sensors such as cameras, Light Detection and Ranging (LIDAR), Radio Detection and Ranging (RADAR) and the Global Positioning System (GPS) to form a representation of the environment. It involves road mapping, vehicle localization, and the detection of static and dynamic obstacles. The decision-making system uses this information to determine appropriate actions for the vehicle. It considers the current state of the vehicle, the environment, and constraints such as traffic regulations, and performs route planning, path planning, obstacle avoidance and vehicle control (Badue et al., 2021).

Recommender systems aim to predict user preferences for products and services, to reduce information overload (Li, 2019). They personalize user experience by understanding and analyzing user behaviour, preferences, and interaction patterns, such as satisfaction, interests, and engagement (Li, 2019). Afsar et al. (2023) state that in traditional recommendation systems, collaborative

filtering methods use patterns of user interactions to identify similarities between users or items, whereas content-based filtering relies on item features and user profiles to match users with relevant content. Modeling the recommendation systems as a sequential decision-making process can better reflect the user-system interaction (Afsar et al., 2023). It is modeled as a Markov decision process (MDP) and uses reinforcement learning algorithms. Reinforcement learning is suited for recommendation systems because it constantly updates its policy based on rewards and accounts for long-term rewards. For recommendation systems, it learns from the user-system interaction, receives feedback as rewards, and maximizes long-term user engagement, satisfaction, and revenue (Afsar et al., 2023).

2. Mathematical Theory

2.1. Markov Decision Process

The Markov Decision Process (MDP) forms the mathematical foundation of reinforcement learning algorithms (Sutton & Barto, 2020). It provides a framework for modeling sequential decision-making problems in stochastic environments (White & White, 1989), where an agent’s actions influence both immediate rewards and subsequent states. These systems satisfy the Markov property, meaning that the state transition and reward depend only on the current state and action, regardless of the prior process (Sutton & Barto, 2020).

According to Sutton and Barto (2020), a MDP is defined by states, actions and rewards, which together describe the interaction between an agent and its environment. The set of all possible states is denoted by $\mathcal{S}$, where each state S_t represents specific coordinates or conditions of the environment at time step t . The agent observes the current state S_t and selects an action A_t , where $\mathcal{A}(s)$ denotes the set of available actions in that state (Sutton & Barto, 2020).

After taking action A_t , the agent transitions to a new state S_{t+1} based on the state transition probability, which represents the likelihood of transitioning to a subsequent state S' given the current state-action pair (Sutton & Barto, 2020):

$$P(S' | s, a) = \Pr(S_{t+1} = S' | S_t = s, A_t = a)$$

Simultaneously, the agent receives an expected immediate reward $R(s, a)$, which is defined as the expected reward received after taking action A_t in state S_t (Sutton & Barto, 2020):

$$R(s, a) = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$$

2.2. Policy and Value Function

A policy, denoted as π , specifies the agent’s behaviour by defining how actions are selected in each state, and $\pi(a | s) = \Pr(A_t = a | S_t = s)$ represents the probability that the agent selects action a when in state s (Sutton & Barto,

2020). The state-value function, $v_\pi(s)$, defines the expected cumulative reward received when the agent starts in state s and follows policy π (Sutton & Barto, 2020). It is defined as :

$$v_\pi(s) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \middle| S_t = s \right]$$

where $\gamma \in [0, 1]$ is the discount factor, and balances immediate reward and the long-term reward. A smaller γ gives more weight to immediate rewards, while a γ closer to 1 gives more weight to long-term rewards.

To evaluate the value of specific actions taken in a state, we use the action-value function (or Q-function), $q_\pi(s, a)$ (Sutton & Barto, 2020). This function measures the expected cumulative reward when the agent starts in state s , takes action a , and follows policy π :

$$q_\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \middle| S_t = s, A_t = a \right]$$

2.3. Bellman Equation

According to Sutton and Barto (2020), the Bellman equation for v_π shows a recursive relationship between the value of a state and the values of subsequent states. It describes the value of a state as the sum of the expected immediate reward and the discounted value of subsequent states. For any policy π , the state-value function satisfies:

$$v_\pi(s) = \sum_a \pi(a | s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_\pi(s')]$$

The Bellman optimality equation can be expressed without reference to a specific policy π , instead selecting the best possible action (Sutton & Barto, 2020). The Bellman optimality equation for the optimal state-value function is:

$$v^*(s) = \max_{a \in \mathcal{A}(s)} \sum_{s', r} p(s', r | s, a) [r + \gamma v^*(s')]$$

It states that the optimal value of a state is the maximum expected total reward achievable by choosing the best action and assuming optimal behaviour in subsequent states. The Bellman optimality equation for the optimal action-value function states the optimal value of a state-action pair:

$$q^*(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} q^*(s', a') \right]$$

It represents the maximum expected total reward obtained by taking action a in state s , followed by optimal behaviour thereafter. It forms the foundation for the Q-learning algorithm (Sutton & Barto, 2020).

2.4. Q-Learning

Sutton and Barto (2020) explain that the model of the environment refers to any mechanism that allows the agent to predict the outcome of its actions. Reinforcement learning methods can be categorized as model-based and model-free, depending on whether the agent has access to an environment model. Model-based agents utilize planning, which is a computational process that uses a model to improve a policy. The agent uses its model to generate simulated experience and evaluate future trajectories. In contrast, model-free agents learn directly from real experience by interacting with the environment, executing actions and observing outcomes (Sutton & Barto, 2020).

According to Sutton and Barto (2020), Q-learning is classified as a model-free, off-policy algorithm. The primary goal of off-policy learning is to evaluate and optimize a target policy, which is intended to be optimal. Simultaneously, the agent follows a distinct, more exploratory behaviour policy to interact with the environment. The ϵ -greedy policy is used in Q-learning to select an action, which balances the trade-off between exploration and exploitation (Sutton & Barto, 2020). With a probability of ϵ , the agent explores by selecting a random (non-greedy) action to improve its estimates of the optimal values. With a probability $1 - \epsilon$, the agent exploits its current knowledge by choosing the greedy action, which is the action with the highest current estimated value (Sutton & Barto, 2020).

Q-learning is a Temporal-Difference (TD) control algorithm that directly approximates the optimal action-value function (Sutton & Barto, 2020). TD error, δ_t , represents the discrepancy between the current estimate of Q-value and the updated estimate of future value from the next state (Sutton & Barto, 2020):

$$\delta_t = R_{t+1} + \gamma Q(S_{t+1}, a') - Q(S_t, A_t)$$

According to Sutton and Barto (2020) and Watkins and Dayan (1992), the Q-learning update rule utilizes this error to improve its estimates:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right]$$

where $\alpha \in [0, 1]$ is the learning rate (or step-size parameter). The Q-value $Q(S_t, A_t)$ is the current estimate of the value of taking action A_t in state S_t . It is updated by incorporating the immediate reward and the discounted value of the best possible action in the subsequent state. This update assumes that the agent follows the optimal policy from the next state onward (Sutton & Barto, 2020; Watkins & Dayan, 1992). A Q-table is a tabular representation of the Q-values, which presents the state-action value estimates (Jang et al., 2019). This table is updated iteratively using the Q-learning update rule, until the values converge to the optimal action-value function $Q^*(s, a)$. Once convergence is achieved, the optimal policy π^* can be obtained by selecting the action that maximizes the value for any given state: $\pi^*(s) = \arg \max_a Q^*(s, a)$ (Sutton & Barto, 2020).

3. Methods

- Christina and Caleb ## Implementation We built and implemented a Q-learning algorithm within a maze environment. The environment was represented as a 6×6 grid constructed using a NumPy array, where integer values encoded state types and associated rewards. Specifically, cells were defined as follows: 0 represented empty traversable space, 1 represented impassable walls, 2 represented fatal states (“holes”) that returned the agent to the starting position, and 3 represented the goal state associated with a positive reward and episode termination.

A Q-table was initialized with zeros to represent the expected value of each state-action pair. The learning rate (α), future discount factor (γ), and exploration rate (ϵ) were predefined and held constant throughout training. The agent’s action space consisted of four discrete movements (up, down, left, right).

At each step, the agent selected an action using the Epsilon-Greedy strategy. State transitions were determined by applying the selected action to the agent’s current position, with resulting coordinates mapped onto the grid to determine the subsequent state and corresponding reward.

Q-values were iteratively updated using the temporal difference learning rule based on the observed reward and the maximum expected future reward from the subsequent state. Training was conducted over a fixed number of episodes, with each episode beginning at the starting state and terminating upon reaching the goal or a terminal penalty state.

3.1. Neuropsychiatry Extension

- Caleb Part
- Q learning in computational psychiatry
- Example: D1/D2 receptors

4. Results

4.1. Model

After running 1000 episodes, we present evidence of the agent’s learning process. Figure 1 illustrates changes in the number of steps the agent takes to complete one episode over the course of 500 episodes.

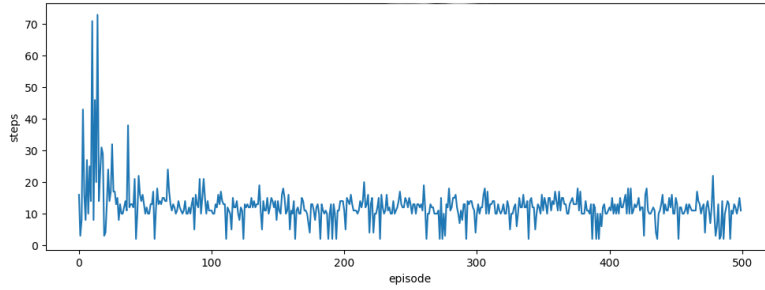


Figure 1: Changes in Steps to Complete One Episode

Figure 2 presents changes in the TD learning error (δ) over time across 6000 episodes.

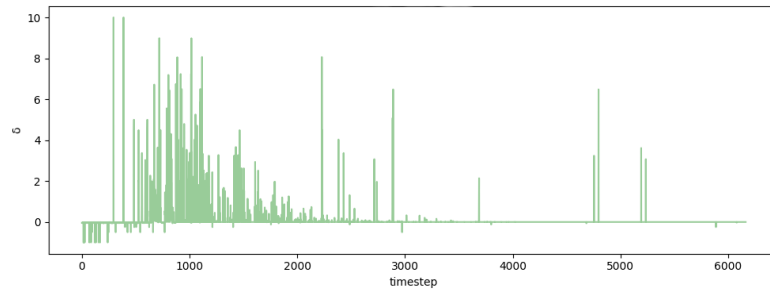


Figure 2: TD Error Over Time

Figure 3 is a Q-value heatmap showing the Q values of the best action to take in a given state (i.e. state-action pair that renders the maximal Q-value), with greater Q-values of darker color.

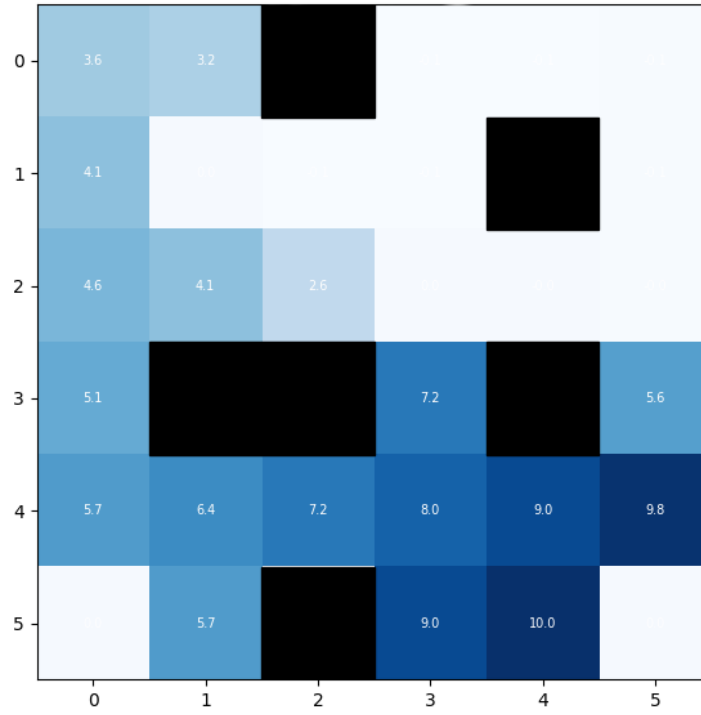


Figure 3: Q-Value Heatmap

Figure 4 illustrates the optimal actions to take in a given state, with visualizations of actions as arrows in the maze environment. This represents the optimal policy the agent is learning.

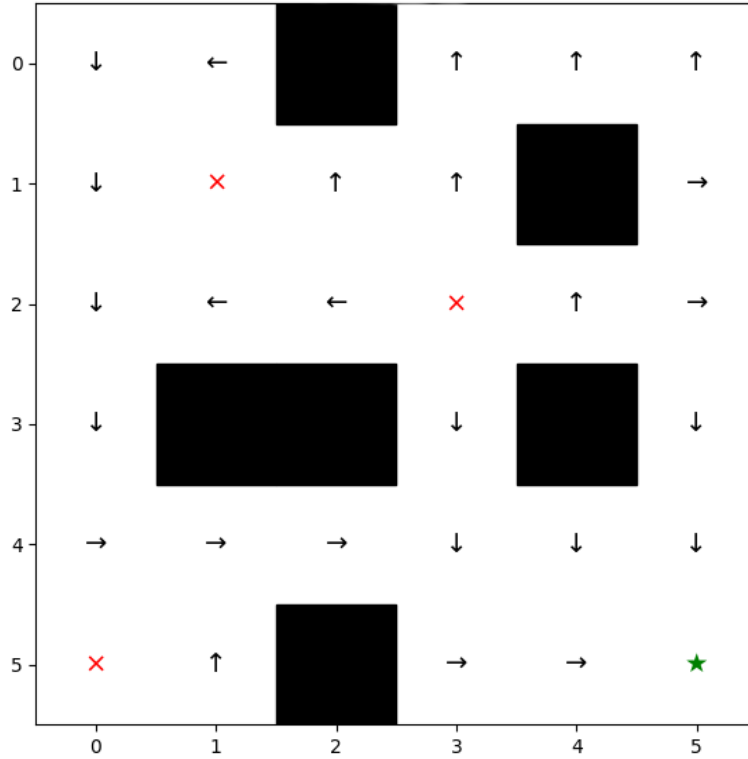


Figure 4: Optimal Learned Policy

4.2. Neuropsychiatry Extension

4.2.1. Dopamine as a Prediction Error Signal

- Schultz et al. (1997)
- DA neurons fire in response to positive prediction error/pause for neg
- This maps onto TD error: biological motivation for using RL as a model for brain function
- D1/D2 receptor mechanisms
- Frank et al. (2004): DA medications in Parkinson's reverse Go/NoGo learning biases/comparison with predictions made with RL models

The application of reinforcement learning algorithms in the study of the brain emerged with advancements in electrophysiological recording techniques in the 1990s. These techniques included single-unit recordings of individual neurons in alert subjects performing behavioural tasks. This allowed the investigation of firing patterns in dopaminergic (DA) neurons in the midbrain, which had been proposed to be responsible for reward-driven learning. Single-unit recordings of DA neurons in the ventral tegmental area (VTA) and substantia nigra of

monkeys performing classical conditioning tasks revealed a unique firing pattern: early in the trials, DA neurons fired phasically at reward delivery; after learning, they fired at the appearance of the cue that predicted reward, not at the reward itself; and when a predicted reward failed to arrive, their firing rate dropped below baseline at exactly the moment the reward would have occurred (Schultz et al., 1997). This suggested that dopaminergic neurons do not simply encode the occurrence of reward, but rather the discrepancy between the predicted and actual timing and magnitude of reward outcomes (Schultz et al., 1997). This discovery motivated a connection to reinforcement learning algorithms, specifically to the temporal difference (TD) error.

Schultz, Dayan, and Montague (1997) interpreted these findings with the TD algorithm, showing that the pattern of DA neuron firing matched the prediction error δ_t (see **Temporal Difference Error**). When the TD model was applied to the same conditioning task that the monkeys performed, the prediction error it generated corresponded precisely to the three observed firing patterns: a positive prediction error when unexpected reward occurred, no error when reward was received as predicted, and a negative prediction error when an expected reward was absent. This established dopamine as a biological implementation of the TD error signal, suggesting that the brain computes something equivalent to TD learning through phasic DA neuron firing activity.

While Schultz et al. (1997) established that phasic dopamine activity encodes reward prediction errors, how this signal drives learning through receptor-level mechanisms remained a question. Specifically, if D1 receptors on the direct pathway facilitate learning from positive outcomes and D2 receptors on the indirect pathway facilitate learning from negative outcomes, then manipulating dopamine levels should selectively affect positive and negative feedback learning in opposite directions. Frank, Seeberger, and O'Reilly (2004) tested this prediction by examining procedural learning in Parkinson's disease patients on and off dopamine medication, using probabilistic selection and transitive inference tasks. Patients off medication — with depleted dopamine — showed enhanced learning from negative feedback but were impaired at learning from positive outcomes, while patients on medication showed the reverse pattern. These results confirmed the predicted double dissociation, establishing that positive and negative feedback learning are mediated by separable dopaminergic mechanisms through D1 and D2 receptors respectively.

4.2.2. Computational Psychiatry

- Huys et al. 2016 - review of comp. psychiatry
- uses computational models to characterize psychiatric conditions mechanistically rather than symptom based
- Theory driven vs. Data driven
- RL parameters as biomarkers
- Q-learning parameters as biomarkers - individual differences in alpha values reflect real biological variation

4.2.3. *Q-Learning in Psychiatric Research*

- Frank et al. 2007
- Introduce their methods
 - probabilistic selection task
 - split alpha into a+ and a-
 - fitting D1/D2 receptors
 - DNA profiles for receptor variants
 - recovering disorder profiles

4.2.4. *Our Simulation*

- replication of Frank et al. (2007) logic in maze environment
- four agents with different alpha values
- learning curves and TD error results
- Interpretation; connect back

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